

Kaushlendra Kumar

Curriculum Vitae

School of Mathematical Sciences
Queen Mary University of London
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Profile

- Research** My research lies in mathematical and theoretical physics, connecting exact gauge fields, quantum/noncommutative geometry, spectral distance, quantum information geometry, and emergent spacetime. Recent work computes black-hole quantum-geometric flow and phase diagnostics, finite spectral distances with quantum-information meaning, exact Yang–Mills/coset fields and source/transport laws, finite Clifford spectral triples, and matrix-model corrections to gravitational dynamics.
- Keywords** Quantum geometry, exact Yang–Mills fields, noncommutative geometry, spectral distance, quantum information geometry, black-hole transport, matrix/emergent gravity.

Positions

- Mar 2024–Nov 2027 **DFG Walter-Benjamin Research Fellow**, *School of Mathematical Sciences*, Queen Mary University of London, Quantum Riemannian Geometry with Shahn Majid.
- Jul 2023–Feb 2024 **Postdoc**, *Institute of Physics*, Humboldt University of Berlin, Germany. Funded by the Volkswagen Stiftung grant of Nichol Furey.
- 2023–2024 (4 months) **Junior Research Fellow**, *Erwin Schrödinger Institute for Mathematics and Physics*, University of Vienna.
- May 2022–Jun 2023 **Postdoc**, *Institute for Theoretical Physics*, Leibniz University Hannover (ITP-LUH), Germany. Funded by DAAD and ITP-LUH.

Major Fellowships

- 2024–2027 **Deutsche Forschungsgemeinschaft (DFG)**, Walter-Benjamin Fellowship, Quantum Riemannian Geometry, Queen Mary University of London.
- 2018–2022 **Deutscher Akademischer Austauschdienst (DAAD)**, Doctoral scholarship, Leibniz University Hannover.
- 2016–2018 **Council for Scientific and Industrial Research (CSIR)**, Junior Research Fellowship (JRF), IISER Kolkata.

Education

- 01 Oct, 2018–04 May, 2022 **PhD Theoretical Physics**, *Institute for Theoretical Physics*, Leibniz University Hannover, Germany.
Thesis title Solutions of Yang–Mills theory in four-dimensional de Sitter space
Grade Magna cum Laude
Defense date 4 May, 2022
- 01 Jun, 2018–30 Sep, 2018 **German language**, *Carl Duisberg Centrum Berlin*, mandatory pre-PhD language course funded by DAAD, Level reached: B1 (Intermediate).
- 23 Jul, 2015–31 May, 2018 **MSc Physics**, *Department of Physical Sciences*, Indian Institute of Science Education & Research, Kolkata (IISER-K), India.
Grade 9.25/10
Defense date 20 Jul, 2017

23 Jul, 2010– **BTech & MTech Biochemical Engineering**, *School of Biochemical Engineering*,
 08 May, 2015 Indian Institute of Technology, Banaras Hindu University (IIT-BHU), Varanasi, India.
 Grade 7.88/10
 Defense date 27 May, 2015

List of publications

(a) Regular articles

- 2024 **Kaushlendra Kumar** and Harold C. Steinacker. Modified Einstein equations from the 1-loop effective action of the IKKT model. *Classical and Quantum Gravity*, volume 41, page 185007, 2024. DOI:10.1088/1361-6382/ad6e4b.
- 2024 Savan Hirpara, **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Exact gauge fields from anti-de Sitter space. *Journal of Mathematical Physics*, volume 65, page 072903, 2024. DOI:10.1063/5.0150027.
- 2022 **Kaushlendra Kumar**, Olaf Lechtenfeld, Gabriel Picanço Costa, and Jona Röhrig. Yang–Mills solutions on Minkowski space via non-compact coset spaces. *Physics Letters B*, volume 835, page 137564, 2022. DOI:10.1016/j.physletb.2022.137564.
- 2022 **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Trajectories of charged particles in knotted electromagnetic fields. *Journal of Physics A: Mathematical and Theoretical*, volume 55, page 315401, 2022. DOI:10.1088/1751-8121/ac7c49.
- 2022 Lukas Hantzko, **Kaushlendra Kumar**, and Gabriel Picanço Costa. Conserved charges for rational electromagnetic knots. *The European Physical Journal Plus*, volume 137, page 407, 2022. DOI:10.1140/epjp/s13360-022-02563-4.
- 2021 **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Instability of cosmic Yang–Mills fields. *Nuclear Physics B*, volume 973, page 115583, 2021. DOI:10.1016/j.nuclphysb.2021.115583.
- 2020 **Kaushlendra Kumar** and Olaf Lechtenfeld. On rational electromagnetic fields. *Physics Letters A*, volume 384, page 126445, 2020. DOI:10.1016/j.physleta.2020.126445.
- 2018 **Kaushlendra Kumar** and Biswajit Chakraborty. Spectral distances on the doubled Moyal plane using Dirac eigenspinors. *Physical Review D*, volume 97, page 086019, 2018. DOI:10.1103/PhysRevD.97.086019.
- 2018 Yendrembam Chaoba Devi, **Kaushlendra Kumar**, Biswajit Chakraborty, and Frederik G. Scholtz. Revisiting Connes’ finite spectral distance on noncommutative spaces: Moyal plane and fuzzy sphere. *International Journal of Geometric Methods in Modern Physics*, volume 15, page 1850204, 2018. DOI:10.1142/S0219887818502043.
- 2015 **Kaushlendra Kumar**, Shivraj Prajapat, and Biswajit Chakraborty. On the role of Schwinger’s SU(2) generators for Simple Harmonic Oscillator in 2D Moyal plane. *The European Physical Journal Plus*, volume 130, page 120, 2015. DOI:10.1140/epjp/i2015-15120-7.

(b) Conference proceedings

- 2023 **Kaushlendra Kumar**. SO(1,3) Yang–Mills solutions on Minkowski space via cosets. *SciPost Physics Proceedings*, volume 14, page 033, 2023. DOI:10.21468/SciPostPhysProc.14.033.
- 2023 **Kaushlendra Kumar**. On Yang–Mills fields from anti-de Sitter spaces. *Journal of Physics: Conference Series*, volume 2667, page 012018, 2023. DOI:10.1088/1742-6596/2667/1/012018.

(c) Preprints

- 2026 **Kaushlendra Kumar** and Shahn Majid. Geometric Aharonov–Bohm phase effect around a black hole. *arXiv preprint*, 2026. arXiv:2605.28582 [gr-qc], preprint.

- 2026 **Kaushlendra Kumar** and Shahn Majid. Geodesic flows on a black-hole background. *arXiv preprint*, 2026. arXiv:2603.03222 [gr-qc], preprint.
- 2026 **Kaushlendra Kumar**. Characteristic Lightcone Sources in $SO(1,3)$ Yang-Mills Theory. *arXiv preprint*, 2026. arXiv:2606.27241 [hep-th], preprint.
- 2026 **Kaushlendra Kumar**. Calibrated Helstrom geometry on the Bloch ball via Connes spectral distance. *arXiv preprint*, 2026. arXiv:2606.13824 [quant-ph], preprint.

Professional service

Referee Journal of Geometry and Physics.

Teaching

- Spring, 2020 **Introduction to Quantum Theory**, *intensive 1-week revision course*, ITP-LUH.
- Fall, 2019 **Mathematical Methods of Physics**, *intensive 1-week revision course*, ITP-LUH.
- Spring, 2019 **Lie groups, Algebra and their Representations**, *teaching assistant*, ITP-LUH.
- Fall, 2017 **Advanced Mathematical Methods of Physics**, *teaching assistant*, IISER-K.
- Spring, 2017 **Intermediate Electricity and Magnetism**, *teaching assistant*, IISER-K.

Talks

Research talks

- 1 Jul, 2026 **XLIII Workshop on Geometric Methods in Physics, Białystok**.
Scalar-anchored Connes distance and Helstrom geometry of the qubit Bloch ball (upcoming conference talk).
- 9 Dec, 2025 **Quantum Algebras seminar, Queen Mary University of London**.
Geodesic flows on a black-hole background.
- 27 Jul, 2023 **QTS12 conference, Prague**.
Exact gauge fields from anti-de Sitter space.
- 4 May, 2023 **Erwin Schrödinger Institute, Vienna**.
Exact gauge fields from anti-de Sitter space.
- 15 Mar, 2023 **Solitons@Work**.
Exact gauge fields from anti-de Sitter space (online).
- 23 Aug, 2022 **String Theory Journal Club, DESY Hamburg**.
Yang–Mills solutions on Minkowski space via non-compact coset spaces.
- 18 Jul, 2022 **GROUP34 conference, Strasbourg**.
Yang–Mills solutions on Minkowski space via non-compact coset spaces.
- 16 Mar, 2022 **Solitons@Work**.
Yang–Mills solutions on Minkowski space via non-compact coset space (online).
- 7 Jul, 2017 **S.N. Bose National Centre for Basic Sciences, Kolkata**.
Spectral distances on doubled Moyal plane.
- 11 Jul, 2016 **S.N. Bose National Centre for Basic Sciences, Kolkata**.
Connes spectral distance on noncommutative spaces: fuzzy sphere & double Moyal plane.

Other seminars and review talks

- 28 May, 2024 **Quantum Algebras seminar, Queen Mary University of London**.
Octonions and the Standard Model.
- 5 Mar, 2024 **Quantum Algebras seminar, Queen Mary University of London**.
Emergence of modified Einstein equation from 1-loop result in IKKT matrix model.

- 18 Aug, 2023 **Balachandran's room 316 meeting (online).**
Emergent gravity from the 1-loop effective action of the IKKT matrix model.
- 28 Jun, 2023 **Lechtenfeld's group seminar, Hannover.**
Emergent gravity from the 1-loop effective action of the IKKT matrix model.
- 1 Jun, 2022 **Lechtenfeld's group seminar, Hannover.**
U(1) gauge theory in noncommutative geometry; pedagogical talk based on Dungen & Suijlekom, Rev.Math.Phys. **24** (2012) 1230004.
- 20 Apr, 2022 **Lechtenfeld's group seminar, Hannover.**
Spectral triple with real structure on fuzzy sphere; review talk based on Chakraborty et. al., J.Math.Phys. **63** (2022) 023504.
- 1 Dec, 2021 **Lechtenfeld's group seminar, Hannover.**
New Yang–Mills solution on positive light-cone.
- 4 Nov, 2016 **DPS journal club IISER, Kolkata.**
Angular momentum & simple harmonic oscillator in noncommutative Moyal plane.

Schools/Conferences

- 15–17 Jul, 2026 **Black Hole Horizons and Beyond**, New Frontiers in Gravity and Holography, Durham, United Kingdom.
upcoming
- 29 Jun–4 Jul, 2026 **XLIII Workshop on Geometric Methods in Physics**, Białystok, Poland.
upcoming; talk on 1 Jul
- 07–11 Jul, 2025 **A Quantum of Noncommutativity**, Workshop/conference in Ambleside, United Kingdom.
- 24–28 Jul, 2023 **QTS12**, The 12th International Symposium on Quantum Theory and Symmetries at Prague, Czech Republic.
- 06–17 Feb, 2023 **Vort2023**, Vortex Moduli workshop at ICTS, Bangalore, India.
online participation
- 29 Aug–9 Sep, 2022 **Saalburg28**, The 28th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Modern methods in quantum field theory" at Bayrischzell, Germany.
- 18–22 Jul, 2022 **GROUP34**, The 34th International Colloquium on Group Theoretical Methods in Physics at Strasbourg, France.
- 30 Aug–10 Sep, 2021 **Saalburg27**, The 27th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Gravitational waves and Black holes" at Heigenbruecken, Germany.
- 02–07 Aug, 2021 **ICMP**, International Congress on Mathematical Physics preceded by Young Researchers Symposium at Geneva, Switzerland.
- 31 Aug–11 Sep, 2020 **Saalburg26**, The 26th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "QFT, Integrable systems and String field theory" at Heigenbruecken, Germany.
- 23–27 Sep, 2019 **SHGP**, Summer School on the topic of "Special Holonomy – Geometry and Physics" at Freiburg, Germany.
- 02–13 Sep, 2019 **Saalburg25**, The 25th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Asymptotic symmetries and Hamiltonian formulation of Gravity" at Heigenbruecken, Germany.
- 15–19 Jan, 2018 **QGNCG**, Conference on "Quantum Groups & Noncommutative Geometry" at National Institute of Science Education & Research (NISER), Bhubaneswar, India.
- 06–24 Nov, 2017 **GGD2018**, Preschool + Advanced school on the topic of "Geometry, Groups and Dynamics" at International Centre for Theoretical Sciences (ICTS) Bangalore, India.

Language skills

Indic Maithili, Hindi and Bengali.
English Fluent.
German Intermediate level (B1).

Research computation

Tools Mathematica/Wolfram Language, $\text{\LaTeX}$, GnuPlot, symbolic and numerical computation.
Methods Finite-dimensional spectral-geometry calculations, exact tensor/curvature computations, Yang–Mills mode algebra, PDE/transport postprocessing.
Systems Linux, macOS, Windows.